Original Article

Morphometric analysis of lumbar pedicles in the Saudi Arabian population – A CT-based study on 1500 vertebrae

ABSTRACT

Background: Knowledge of the lumbar pedicle anatomy is vital for preoperative surgical planning of pedicle screw fixation procedures in orthopedic surgery. The morphology of lumbar vertebrae has both genetic and ethnic variations.

Study Design: Retrospective cohort study.

Objective: The objective of the study was to estimate morphometric characteristics of the lumbar vertebrae relevant for pedicle screw placements in the Saudi population and compare it with CT-based studies in other population groups.

Materials and Methods: CT scans of 300 Saudi patients (M = 150; F = 150) were reviewed. The dimensions of the lumbar pedicle, pedicle axis angle, and chord length (CL) of the L1–L5 vertebrae were measured by two investigators. The Mann–Whitney test was used to compare the genders.

Results: There was no inter-observer bias. There was no difference between the sides. Transverse pedicle diameter (TPD) increased from L1 to L5 with mean values of 5.76–13.62 mm, respectively. The longest CL was at L3 with a mean of 50.92 mm. The length of the pedicle decreased from L1 to L5 with mean values of 16.01–9.93 mm, respectively. The height of the pedicle (PH) showed a similar trend with a decrease from 9.75 to 8.3 mm. The pedicle axis angle trajectory followed a gradual medial angulation pattern of 12.68°–28.23° from L1 to L5.

Conclusions: The TPD, CL, and PH showed statistically significant differences among the genders. Statistical significance was also noted among pedicle parameters compared with other population groups.

Keywords: Computed tomography, lumbar vertebra, pedicle, spine

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INTRODUCTION

The lumbar spine is composed of five large vertebrae whose main adaptation is to bear weight. It is composed of a large reniform body anteriorly and a posterior vertebral arch that is attached to the body through the pedicle. Pedicle screw instrumentation is the most widely accepted method globally to provide mechanical stability to the lumbar spine and is used in the treatment of fractures, spondylolisthesis, tumors, scoliosis, kyphosis, and other degenerative disorders.[1] Degenerative diseases are often treated by fusion of the spine supplemented by the use of bilateral or unilateral pedicle screw fixation.[2] Insertion of pedicle screws involves associated risks to adjacent neurovascular structures like lumbar nerve roots, meninges, the aorta, inferior vena cava,

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and abdominal visceral organs. Appropriate three-dimensional knowledge of the morphology of vertebrae is of paramount importance to all spine surgeons for careful selection and safe insertion of pedicle screws. Thus, pedicle parameters play a crucial role in the sizing and trajectory of the pedicle screws.

The morphology of the vertebrae is known to exhibit genetic and ethnic variations.[3-5] Multiple anatomical and radiological studies have confirmed the variations in morphology of the vertebrae in different populations and ethnic groups.[6-11] There is a paucity of data from the Middle Eastern population regarding the anatomy of the lumbar spine with particular reference to pedicle screw fixation. This study aimed to develop a gender-based database of the morphometric parameters of the L1–L5 lumbar vertebrae representative of the Saudi Arabian population with particular reference to pedicle screw stabilization procedures. The primary objective was to propose ideal screw trajectories, screw dimensions, and safe bony corridors for the placement of pedicle screws in the Saudi population. Our second objective was to compare the bony parameters and proposed screw techniques with computed tomography (CT) studies done in other populations or racial groups.

MATERIALS AND METHODS

The study was conducted after obtaining ethical approval from the Institutional Review Board (IRB no. 19-563). CT scans of all adult patients (aged 18–60 years) who had undergone imaging of the abdomen or lumbar spine for suspected abdominal pathology or low back ache during the period of 5 years (2015–2020) were reviewed retrospectively. All patients who had congenital or acquired deformities, fractures, spondylolysis or spondylolisthesis, or a history of previous spine surgery (instrumented or noninstrumented) were excluded from the study. De-identified data were obtained and entered into an Excel sheet and stored in password-protected computers.

Measured parameters

The morphometry of the pedicle, pedicle-axis angle, and chord length (CL) was measured in all lumbar vertebrae, i.e. L1–L5, as described by Vaccaro and Garfin[12]

- Transverse pedicle diameter (TPD) – the distance between the medial and lateral outer cortical width of the pedicle at the level of the isthmus [Figure 1]
- Length of the pedicle (PL) – the distance from the posterior cortex of the pedicle to the posterior longitudinal ligament in line with the mid-axis of the pedicle [Figure 2]
- Height of the pedicle (PH) – the distance between the superior and inferior cortex in the sagittal plane at the level of the isthmus [Figure 3]
- CL – the distance measured from the posterior cortical entry point of the pedicle to the anterior vertebral cortex in line with the mid-axis of the pedicle [Figure 1]
- Pedicular axis angle – the angle formed by the mid-axis of the pedicle to the midline [Figure 2].

The results were stratified based on the side of the vertebra and gender, and a Student *t*-test was employed to analyze for any significant difference. The results were also compared with

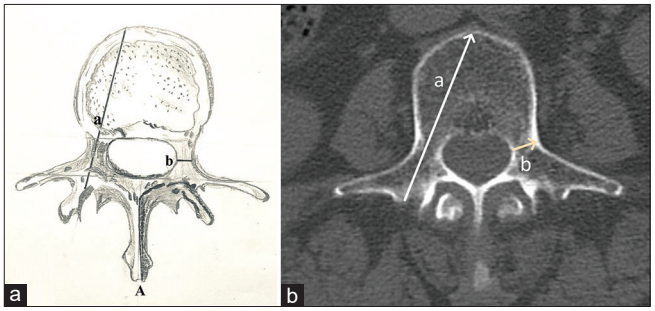


Figure 1: (A) An axial illustration of the lumbar vertebra; (B) computer tomographic image of a lumbar vertebra showing (a) chord length (the distance measured from the posterior cortical entry point of the pedicle to the anterior vertebral cortex in line with the mid-axis of the pedicle) and (b) transverse pedicle diameter (the distance between the medial and lateral outer cortical width of the pedicle at the level of the isthmus)

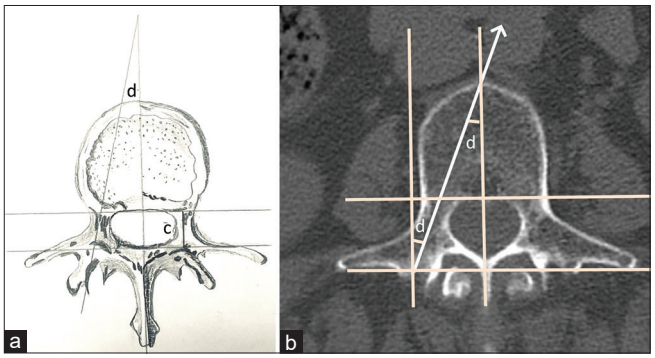


Figure 2: (A) An axial illustration of the lumbar vertebra; (B) computer tomographic image of a lumbar vertebra showing; (c) length of the pedicle (distance from the posterior cortex of the pedicle to the posterior longitudinal ligament in line with the mid-axis of the pedicle); (d) pedicular axis angle (the angle formed by the mid axis of the pedicle to the midline)

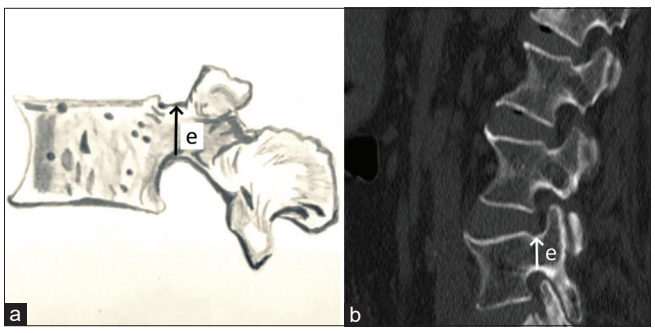


Figure 3: (A) Sagittal illustration of the lumbar vertebra; (B) computer tomographic image of a lumbar vertebra showing; (e) height of the pedicle

other CT-based morphometric studies in other population groups using the Student *t*-test. Only CT-based studies that adequately presented data were included. All case reports, plain radiographic measurements, manual measurements in anatomical specimens, or intraoperative measurements, and studies involving children were excluded.

Imaging protocol and imaging postprocessing

Noncontrast CT images extending from the first lumbar to fifth sacral vertebrae were taken in 2.5 mm contiguous axial sections with the following imaging parameters: 140 kV, 500 mA, 0.7 s rotation time, number of images being 200, total exposure time of 5 s, and the dose length product average between 700 and 900 mGycm.

Raw data sets were loaded into a dedicated diagnostic workstation (AW 5, GE Healthcare Technologies, WI) equipped with a professional postprocessing software package (Centricity Enterprise zero footprint software – GE healthcare system USA version: 6.0 SP4.

Sample size and statistical analysis

A study of the TPD by Dzierżanowski *et al*., on 100 subjects, reported the mean to be 6.1 ± 1.5 mm.[10] Assuming a precision of 0.4 and a confidence interval of 95%, the minimum sample size required is 55 subjects. As the research also observed gender differences, a total of 150 males and 150 females were studied (Total = 300 subjects) by consecutive sampling. All parameters were measured individually by two radiologists with equal experience. All parameters were measured three times, and the average value was taken.

The data were analyzed using SPSS v22 (IBM, Chicago, Illinois, USA). The Student *t*-test was used to compare the parameters in both genders. Mean, range, and standard deviation were estimated. A $P < 0.05$ was considered statistically significant. Inter-observer reliability test was also calculated to estimate the reliability of measured values between the investigators. Mean of the two population groups were compared using the Student *t*-test to observe variation in the measured parameters.

RESULTS

A total of 300 patients (males = 150, and females = 150) fulfilled the inclusion criteria and were enrolled in the study. The average age was 42.0 years (range from 18.0 to 60.0). All parameters were measured in each of the lumbar vertebrae from L1 to L5. Inter-rater variability analysis revealed that there was no interobserver bias ($P > 0.05$). The side-specific pedicle parameters are shown in Table 1. No statistical significance was noticed in the measured parameters between the sides of the vertebra. Table 2 shows the gender-based comparison of measured parameters. Table 3 and Figure 4 compare the mean values of the Saudi study with other populations using the Student *t*-test for significance.

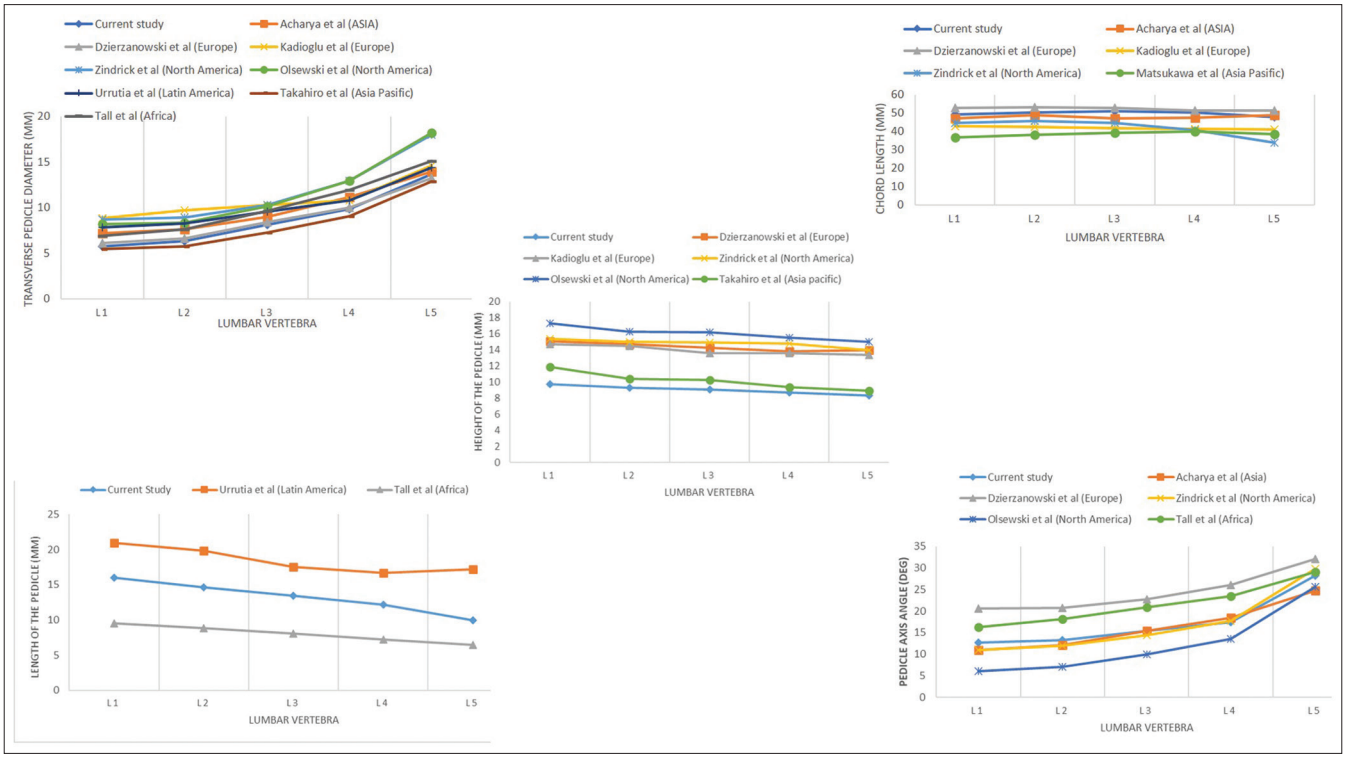


Figure 4: Cumulative graphs of all pedicle parameters comparing the current Saudi study with other population groups

Table 1: Comparison of lumbar pedicle parameters (mean±standard deviation) with respect to the distinguished side across gender

Lumbar vertebra	Laterality	Female	Male	Mean	*P*	Correlation
TPD (mm)						
L1	Left	5.4±0.9	6.2±1	5.8±1.1	<0.001	0.066
	Right	5.3±0.9	6.1±1.1	5.7±1.1	<0.001	0.048
L2	Left	6±0.9	6.7±0.9	6.3±1	<0.001	0.035
	Right	5.9±1	6.7±0.9	6.3±1	<0.001	0.010
L3	Left	7.6±1	8.6±1	8.1±1.1	<0.001	0.047
	Right	7.5±1.1	8.5±1	8±1.1	<0.001	0.071
L4	Left	9.4±1.2	10.4±1.2	9.9±1.3	<0.001	0.103
	Right	9.3±1.1	10.2±1	9.7±1.2	<0.001	0.062
L5	Left	13.3±1.5	14±1.5	13.6±1.5	<0.001	0.039
	Right	13.3±1.5	13.9±1.5	13.6±1.5	0.002	0.057
CL (mm)						
L1	Left	47.6±2.9	50.7±3.3	49.1±3.4	<0.001	0.057
	Right	47.8±2.9	50.6±3	49.2±3.3	<0.001	0.068
L2	Left	48.8±2.4	51.5±3.2	50.2±3.1	<0.001	0.119
	Right	49.2±2.6	52±3.1	50.6±3.2	<0.001	0.132
L3	Left	49.5±2.5	51.7±2.9	50.6±2.9	<0.001	0.039
	Right	50.1±2.5	52.3±2.8	51.2±2.9	<0.001	0.041
L4	Left	48.9±2.9	51.1±3.6	50±3.4	<0.001	0.075
	Right	49.2±2.9	51.4±3.5	50.3±3.3	<0.001	0.057
L5	Left	46.4±3.3	48.8±3.4	47.6±3.6	<0.001	0.134
	Right	46.3±3.1	48.9±3.6	47.6±3.6	<0.001	0.079
Pedicle length (mm)						
L1	Left	15.6±1.2	16.3±1.3	16±1.3	<0.001	0.098
	Right	15.7±1.3	16.3±1.3	16±1.3	<0.001	0.090
L2	Left	14.4±1.2	14.6±1.2	14.5±1.2	0.122	0.042
	Right	14.6±1.1	14.8±1.2	14.7±1.2	0.165	0.045
L3	Left	13.3±1.2	13.4±1.2	13.3±1.2	0.510	0.112
	Right	13.4±1.2	13.5±1.2	13.5±1.2	0.498	0.069
L4	Left	12±1.2	12.1±1.2	12.1±1.2	0.529	0.024
	Right	12.2±1.1	12.3±1.1	12.2±1.1	0.330	-0.001
L5	Left	9.7±1.3	9.9±1.3	9.8±1.3	0.070	0.009
	Right	9.9±1.4	10.1±1.4	10±1.4	0.307	0.013
Axis of pedicle (°)						
L1	Left	12.8±2.8	13.7±2.6	13.3±2.7	0.004	0.002
	Right	11.7±2.6	12.3±2.7	12±2.7	0.042	0.075
L2	Left	13.6±2.4	14.1±2.6	13.8±2.5	0.104	0.043
	Right	12.4±2.2	13±2.5	12.7±2.4	0.025	0.066
L3	Left	15.9±2.6	16.4±2.8	16.2±2.7	0.108	-0.004
	Right	14.6±2.3	14.9±2.5	14.7±2.4	0.239	0.096
L4	Left	17.9±3.4	18.1±3.5	18±3.4	0.506	0.086
	Right	16.6±3.4	17.1±3.4	16.9±3.4	0.209	0.143
L5	Left	29.3±4.9	29.5±3.7	29.4±4.3	0.738	-0.013
	Right	26.8±4.2	27.2±3.4	27±3.8	0.433	-0.018
Height of pedicle (mm)						
L1	Left	9.2±1.2	10.1±1.1	9.7±1.2	<0.001	0.116
	Right	9.4±1.3	10.2±1.2	9.8±1.3	<0.001	0.130
L2	Left	9.1±1.2	9.6±1.1	9.3±1.2	<0.001	0.091
	Right	8.9±1.3	9.6±1	9.3±1.2	<0.001	0.120
L3	Left	8.9±1.2	9.4±1	9.2±1.1	<0.001	0.120
	Right	8.7±1.2	9.3±1.1	9±1.2	<0.001	0.141
L4	Left	8.5±1.2	9±0.9	8.8±1.1	<0.001	0.053
	Right	8.3±1.1	8.9±1	8.6±1.1	<0.001	0.095
L5	Left	7.9±1	8.5±0.9	8.2±1	<0.001	0.084
	Right	8.2±1.1	8.5±1	8.3±1	0.001	0.056

TPD - Transverse pedicle diameter; CL - Chord length

Table 2: Comparison of pedicle parameters (mean±standard deviation) with respect to the lumbar vertebra across gender

Lumbar vertebra	Female	Male	Mean	*P*
TPD (mm)				
L1	5.33±0.88	6.19±0.99	5.76±1.02	<0.001
L2	5.95±0.9	6.72±0.87	6.34±0.96	<0.001
L3	7.61±0.98	8.6±0.99	8.11±1.1	<0.001
L4	9.33±1.12	10.3±1.08	9.82±1.2	<0.001
L5	13.3±1.48	13.94±1.42	13.62±1.48	<0.001
CL (mm)				
L1	47.69±2.84	50.68±3.02	49.18±3.29	<0.001
L2	49±2.39	51.78±3.08	50.39±3.09	<0.001
L3	49.81±2.46	52.02±2.82	50.92±2.86	<0.001
L4	49.07±2.78	51.24±3.45	50.15±3.31	<0.001
L5	46.38±3.04	48.89±3.4	47.63±3.45	<0.001
Pedicle length (mm)				
L1	15.69±1.24	16.33±1.23	16.01±1.27	<0.001
L2	14.53±1.11	14.73±1.2	14.63±1.16	0.138
L3	13.38±1.17	13.47±1.18	13.42±1.17	0.509
L4	12.12±1.12	12.23±1.13	12.18±1.12	0.407
L5	9.82±1.31	10.04±1.3	9.93±1.31	0.145
Axis of pedicle (°)				
L1	12.3±2.55	13.06±2.47	12.68±2.54	0.009
L2	12.99±2.16	13.53±2.41	13.26±2.3	0.042
L3	15.26±2.29	15.67±2.43	15.47±2.37	0.131
L4	17.28±3.27	17.66±3.32	17.47±3.3	0.322
L5	28.11±4.38	28.36±3.29	28.23±3.87	0.57
Height of pedicle (mm)				
L1	9.34±1.15	10.16±1.09	9.75±1.2	<0.001
L2	9.01±1.16	9.62±1	9.32±1.12	<0.001
L3	8.82±1.16	9.41±0.96	9.11±1.1	<0.001
L4	8.46±1.08	8.96±0.9	8.71±1.03	<0.001
L5	8.05±1	8.55±0.9	8.3±0.99	<0.001

TPD - Transverse pedicle diameter; CL - Chord length

Transverse pedicle diameter

Our study shows that there was a gradual increase in diameter from L1 to L5. The largest TPD was at L5 with a mean of 13.62 ± 1.48 mm (10.66–16.58) and the smallest at L1 with a mean of 5.76 ± 1.02 mm (3.72–7.80). The TPD at each of the vertebral levels was significantly larger in the male subjects ($P < 0.001$).

Length of the pedicle

The longest pedicle was at the level of the L1 vertebra with a mean of 16.01 ± 1.27 mm (13.47–18.55 mm), and the smallest pedicle length was at the L5 vertebra with a mean of 9.93 ± 1.31 mm (7.31–12.55). There was no statistical significance between the genders in relation to the PL, except at the level of the L1 vertebra ($P < 0.001$).

Height of the pedicle

The tallest pedicle was observed at L1 vertebra with a mean of 9.75 ± 1.2 mm (7.35–12.15 mm), and the shortest pedicle was at the L5 vertebra with a mean of 8.3 ± 0.99 mm (range, 6.32–10.28 mm). There was statistical significance among the genders ($P < 0.001$) in relation to the PH. Pedicle heights were significantly smaller in females than in males.

Chord length

A Caret (inverted v-shaped grapheme) pattern was noticed with regard to the CL. There was a gradual increase in the length from L1 to L3 (49.18 mm to 50.92 mm) followed by a decrease in the CL from L3 to L5 (50.92 to 47.63 mm). The longest CL was noticed at L3 with a mean of 50.92 ± 2.86 mm (range, 45.2–56.64). There was a statistically significant difference seen among the genders with shorter CL for females as compared to the males ($P = 0.001$).

Pedicle axis angle

The angle of the pedicle, described as the angle between the mid-pedicle axis and the midline showed a gradual increase from L1 to L5. The least was at L1 with a mean of 12.68° ± 2.54° (range, 7.6°–17.06°), and the most medially angulated pedicle was at L5 vertebra with a mean of 28.23° ± 3.87° (range, 20.49°–35.97°).

There was no statistically significant difference between the genders ($P < 0.05$) only at the L1 and L2 vertebrae. The angle of medial inclination gradually increased by 2° in each vertebra from L1 to L4, but showed a sudden large increase of 11° from L4 to L5. A medial angulation of 10° at the L1 vertebra and 25° at the L5 vertebra is safe without any medial pedicle breach.

DISCUSSION

This study presents a detailed morphometric analysis of adult lumbar pedicles (L1–L5) obtained exclusively with CT imaging in a large subset of patients treated at a tertiary care center in Riyadh, Saudi Arabia, which is representative of the Saudi population.

This study also compared the data obtained in the Saudi population with exclusively CT-based morphometric studies of lumbar pedicles in other population groups [Table 3]. This showed a significant difference in the measured parameters between the Saudi population and other population-based studies, except for the pedicle axis angle at the L3 vertebra in Acharya's study in the Indian population [Table 3].[13] The findings of this study carry huge clinical relevance as bone mineral density, insertion depth of pedicle screw and insertion angle bear a significant relationship to screw pull-out strength.[14]

Transverse pedicle diameter

TPD is an important parameter to decide the width of the pedicle screw instrumentation without any mediolateral

Table 3: Intercontinental and regional analysis of pedicle parameters (mean±standard deviation) with respect to the lumbar vertebrae exclusively based on computed tomography studies

Parameters	Continents/regions	Author	L1	L2	L3	L4	L5
TPD	Current study	Arockiaraj *et al*.[23] (*n*=300)	5.76±1.02	6.34±0.96	8.11±1.1	9.82±1.2	13.62±1.48
	Asia	Acharya *et al*.[13] (*n*=50)	7.20±0.9*	7.62±0.8*	8.97±1.1*	11.12±1.0*	13.91±1.2*
	Europe	Dzierzanowski *et al*.[10] (*n*=100)	6.1±1.5*	6.6±1.8*	8.4±2.1*	10±2*	13.2±2.5*
		Kadioglu *et al*.[22] (*n*=29)	8.8±0.4*	9.7±2.0*	10.3±2.0*	10.8±2.5*	14.6±3.8*
	North America	Zindrick *et al*.[25] (*n*=26-56)	8.7±2.3*	8.9±2.2*	10.3±2.61*	12.9±2.1*	18.0±4.1*
		Olsewski *et al*.[11] (*n*=39-42)	8.2±2*	8.3±1.9*	10.1±2.2*	12.9±2.4*	18.2±3.1*
	Latin America Region	Urrutia *et al*.[21] (*n*=60)	7.81±1.30*	8.26±1.42*	9.57±0.99*	10.79±0.59*	14.36±1.85*
	Asia Pacific Region	Makino Takahiro *et al*.[24] (*n*=50)	5.46±1.69*	5.76±1.73*	7.25±1.82*	9.01±1.91*	12.86±2.47*
	Africa	Tall *et al*.[23] (*n*=70)	6.9	7.6	9.6	11.9	15.1
Pedicle height	Current study	Arockiaraj *et al*.[23] (*n*=300)	9.75±1.2	9.32±1.12	9.11±1.1	8.71±1.03	8.3±0.99
	Europe	Dzierzanowski *et al*.[10] (*n*=100)	15.1±1.1*	14.7±1.2*	14.3±1.2*	13.8±1.4*	14±1.1*
		Kadioglu *et al*.[22] (*n*=29)	14.7±1.7*	14.5±2.4*	13.6±1.6*	13.6±1.8*	13.4±1.7*
	North America	Zindrick *et al*.[25] (*n*=21–52)	15.4±2.8*	15.0±1.5*	14.9±2.4*	14.8±2.1*	14.0±2.3*
		Olsewski *et al*.[11] (*n*=37–42)	17.3±1.7*	16.3±1.6*	16.2±1.5*	15.5±1.8*	15±2*
	Asia Pacific Region	Takahiro *et al*.[24] (*n*=50)	11.89±1.57*	10.44±1.75*	10.23±1.63*	9.36±1.36*	8.95±1.54*
Pedicle length	Current study	Arockiaraj *et al*. (*n*=300)	16.01±1.27	14.63±1.16	13.42±1.17	12.18±1.12	9.93±1.31
	Latin America Region	Urrutia *et al*.[21] (*n*=60)	20.92±2.62*	19.86±2.93*	17.57±2.12*	16.67±2.13*	17.23±1.35*
	Africa	Tall *et al*.[23] (*n*=70)	9.5	8.8	8.1	7.25	6.45
Pedicle axis	Current study	Arockiaraj *et al*.[23] (*n*=300)	12.68±2.54	13.26±2.3	15.47±2.37	17.47±3.3	28.23±3.87
	Asia	Acharya *et al*.[13] (*n*=50)	10.9±3.1*	12.12±2.9*	15.4±3.6	18.4±4.3*	24.7±3.8*
	Europe	Dzierzanowski *et al*.[10] (*n*=100)	20.6±2.8*	20.7±2.4*	22.8±2.6*	26.1±3*	32.1±4.2*
	North America	Zindrick *et al*.[25] (*n*=26–76)	10.9±2.2*	12.0±3.5*	14.4±3.8*	17.7±5.2*	29.8±6.3*
		Olsewski *et al*.[11] (*n*=41–42)	6±2*	7±3.2*	10±3.5*	13.5±5*	25.6±5.5*
	Africa	Tall *et al*.[11] (*n*=70)	16.3	18.2	20.8	23.5	29.1
CL	Current study	Arockiaraj *et al*.[23] (*n*=300)	49.18±3.29	50.39±3.09	50.92±2.86	50.15±3.31	47.63±3.45
	Asia	Acharya *et al*.[13] (*n*=50)	47.0±3.4*	49.0±2.8*	47.2±3.9*	47.5±5.4*	48.9±4.4*
	Europe	Dzierzanowski *et al*.[10] (*n*=100)	52.9±3.7*	53.1±4*	52.8±4.2*	51.2±3.8*	51.2±4.6*
		Kadioglu *et al*.[22] (*n*=29)	42.7±2.7*	42.5±3.4*	41.6±3.1*	41.3±2.1*	40.8±3.0*
	North America	Zindrick *et al*.[25] (*n*=26–74)	44.7±4.5*	45.5±3.7*	44.4±5.1*	40.7±4.3*	33.7±5.6*
	Asia Pacific region	Matsukawa *et al*.[29] (*n*=100)	36.8	38.2	39.3	39.8	38.3

*Statistical significance. TPD - Transverse pedicle diameter; CL - Chord length

breach. Our study shows a gradual increase in diameter from L1 to L5. The largest TPD was at L5 with a mean of 13.62 ± 1.48 mm (10.66–16.58), and the smallest at L1 with a mean of 5.76 ± 1.02 mm (3.72–7.80). The results imply that the L1 vertebra can accommodate 4.0–5.0 mm pedicle screws, while the rest of the lumbar vertebrae can easily accommodate 6.0 mm or 6.5 mm pedicle screws. Pedicle fracture or pedicle cut-out is possible when more than 80% of the outer pedicle diameter is violated.[15,16] Our study shows a gradual increase in the TPD from L1 to L5 which follows a similar trend as observed in other studies from different population groups – Asians,[13,17,18] North Americans,[19,20] Latin Americans,[16,21] Europeans,[10,22] and Africans[23] [Figure 4]. Studies done among the Caucasian populations of the North American and European regions demonstrate the largest TPD, whereas Asians, including the Saudi population, have shown the least TPD [Table 3]. Makino *et al*., studied an Asian subset and reported the smallest TPD at both L1 and L5 among the available literature.[24] Olsewski *et al*.[11] and Zindrick *et al*.[25] reported the largest TPD (8 mm at L1 and 18 mm at L5) in the North American population group [Table 3]. Asian population was found to have smaller pedicle diameter than the Western population, which is similar to the analysis done by Tse *et al*.[26] and Albano *et al*.,[16] The above findings may be correlated with the fact that the Caucasian population are generally much taller individuals when compared to individuals from the Asia Pacific region. Taller individuals are known to accommodate larger pedicle screws than the shorter individuals.[27] Statistical significance was seen among all the continents and regions when compared to the current study [Table 3].

Chord length

Biomechanical evaluation of the insertion depth of the pedicle screw revealed that increased screw depth resulted in increased fixation, less loosening, and increased pull-out

strength during cyclic loading only in the mid-body and bi-cortical purchase.[28] Furthermore, biomechanically a minimum of 50%–80% of the vertebral body purchased by the pedicle screw makes the construct sound. Matsukawa *et al*., in their study on the Japanese population (Asia-Pacific region), reported the smallest CL (average of 38.5 mm) at all the lumbar vertebral levels when compared to the Caucasians and Asian population.[29] Dzierżanowski *et al*., reported the highest CL of approximately 52.0 mm in the European population.[10] Our current study in the Saudi population reports one of the highest mean CL of approximately 50.0 mm. Surprisingly, Zindrick *et al*. reports the shortest CL 33.7 ± 5.6 mm in the North American population at the L5 vertebral [Figure 4].[25] CL is of paramount importance for both good vertebral body purchase and to prevent inadvertent anterior breach, which may cause damage to major vessels. CL has a statistically significant impact on resistance to lateral bending, axial rotation loading ($P < 0.01$), and mechanical stress at the bone-screw interface ($P < 0.01$).[30]

Length and height of the pedicle

There was a gradual reduction in the height [Figure 4] of the lumbar pedicle from Lumbar 1 to Lumbar 5 vertebra. This was similar to the reports from several researchers among various population groups, such as Urrutia *et al*.,[21] Dzierżanowski *et al*.,[10] Kadioglu *et al*.[22] Tall *et al*.[23] and Zindrick *et al*.[25] [Table 3]. Caucasians had the greatest height of lumbar pedicle whereas the Saudis have the least. Very few researchers have studied the PL of which Caucasians had the highest length.

Pedicle axis angle

Pedicle axis angle followed a similar trend of increase in the medial angle from Lumbar 1 (12.7°) to Lumbar 5 (28.2°) in the Saudi population. A statistically significant difference was not noticed between the genders ($P < 0.05$) only at the L1 and L2 vertebrae. The angle of medial inclination gradually increased by 2° in each vertebra from L1 to L4 but showed a sudden large increase of 11° from L4 to L5. A medial angulation of 10° at L1 vertebra and 25° at the L5 vertebra is safe without any medial pedicle breach. Similar reports of gradual increase are found in most of the population groups [Table 3]. However, a sudden significant increase in medial angulation from the L4 to L5 vertebra has been reported in North American and North Indian studies.[9,11,25] Olsewski *et al*.,[11] reported the least angulation of 6° at L1 and 20° at L5 while Dzierżanowski *et al*.,[10] reported the highest angulation of 20° at L1 and 32° at L5.

This study provides a detailed description of the various pedicle parameters of the lumbar spine in the Saudi Arabian population from a tertiary care center that caters to patients from almost all parts of the country. The baseline data of morphometric parameters of the lumbar pedicle and safe screw trajectory will certainly prepare the spine surgeon for a better understanding of the anatomy of the lumbar spine in this population and suitable preoperative planning. Screw selection based on the database will aid in better pedicle orientation and precise placement of the screws, thereby minimizing the risk of complications. Such a database will also help in population-based designing of pedicle screws and paves the way for further research in this field.

CONCLUSIONS

This study provides a detailed morphometric analysis of the lumbar vertebra with specific relevance to pedicle screw fixation in the Saudi Arabian population. Although all pedicle parameters were generally larger in Saudi males compared to females, statistical significance was seen in the measured values of TPD, CL, and height of the lumbar pedicles among the genders. A significant difference was also noted in pedicle parameters when compared with population-based studies in the North American, European, Latin American, and Asia Pacific regions. This study provides a detailed comparison of pedicle dimensions and screw trajectories in the Saudi population. Its comparison with available literature in different population groups will assist surgeons to maximize surgical outcomes in pedicle screw instrumentation in diverse population settings.

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Conflicts of interest

There are no conflicts of interest.

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